

PRODIPTA SEN AMARTYA

☎ +8801517-006253 • ✉ prodiptasen-2018325324@cs.du.ac.bd
in www.linkedin.com/in/prodipta-sen-amartya • 🌐 <https://github.com/Prodipta0109>

EDUCATION

University of Dhaka • Dhaka, Bangladesh January 2019 – December 2023
Bachelor of Science • Computer Science and Engineering • CGPA: 3.49/4.0

Test of English as a Foreign Language (ToEFL) • November 2024
Overall: 95 (Reading: 25 • Listening: 22 • Speaking: 24 • Writing: 24)

PUBLICATIONS

- **Prodipta Sen Amartya**, Shaily Kabir, Sagar Chandra Karmaker Babu, and Mosarrat Jahan. “An Interval Creation Approach to Construct Interval Type-2 Fuzzy Sets.” Published in the *2024 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, held as part of the *2024 IEEE World Congress on Computational Intelligence (WCCI)*. ([Paper-link](#))

PRESENTATIONS

- Presented “An Interval Creation Approach to Construct Interval Type-2 Fuzzy Sets” in the session ‘Virtual: Foundations of Fuzzy Systems’ at the *2024 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, held as part of the *2024 IEEE World Congress on Computational Intelligence (WCCI)*, on June 30, 2024. ([Presentation slide](#))

RESEARCH ACTIVITIES

Evaluation of the Behaviour and Performance of Interval Type-2 Fuzzy Set
University of Dhaka, Dhaka, Bangladesh March 2023 – January 2024

- Investigated the comparative analysis on efficiency among type-1, general type-2 and interval type-2 fuzzy set
- Designed the whole methodology (Interval Creation Approach - ICA) to construct respective interval type-2 fuzzy set from different types of type-1 fuzzy set (triangular, trapezoidal, Gaussian, generalized bell-shaped curve, symmetric and asymmetric)
- Made the work-plan to demonstrate the effectiveness of ICA with synthetic and real-world datasets to prove its competency
- Performed a detailed inspection contrasting the performance of the proposed ICA method with regard to the Interval Approach - IA, Enhanced Interval Approach - EIA, and M2 methods; previously proposed methods to convert type-1 fuzzy set into interval type-2 fuzzy set
- Reduced the extra pre-processing time; thus improved the overall computational cost
- Contributed as the first author of the paper named "[An Interval Creation Approach to Construct Interval Type-2 Fuzzy Sets](#)" based on this research
- **Technical Skills:** Python, Java, C++, Matlab, PYIT2FLS, Juzzy, collaborative and content-based filtering, neural networks, evolutionary algorithms
- **Soft Skills:** Team leadership, planning, presentation skills, paper and report writing

Ultrasound Technology for Improvement of 3D Imaging
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh January 2024 – Present

- Analyzing the characteristics of ultrasound to improve the quality of the transducer to capture better 3D image

- Developing an optimized technique to detect and classify breast tumour-affected lesion with help of ultrasound video stream
- **Technical Skills:** Python, computer vision techniques
- **Soft Skills:** Team-collaboration, punctuality, critical thinking

Sensitivity Analysis with Fuzzy Sets

Supervisor: Dr. Shaily Kabir, LUCID Lab, University of Nottingham, UK

- Conducting research to identify which input variables or fuzzy rules can ensure robustness and reliability of a fuzzy inference system
- Determining the effectiveness of non-singleton fuzzy set in respect to level of noise in the dataset
- **Technical Skills:** FuzzyR, explainable artificial intelligence
- **Soft Skills:** rigorous analysis, communication

Security Analysis in Federated Learning and Split Learning

Supervisor: Dr. Md Tariqul Islam,

University of Maryland - Baltimore County, USA

July 2025 – February 2026

- Transferability of Attack and Defense between Federated Learning and Split Learning
- **Technical Skills:** Federated Learning (FedAvg, FedProx), Split Learning architectures, adversarial attacks (model inversion, membership inference, gradient leakage), defense mechanisms (differential privacy, secure aggregation), explainable artificial intelligence, Python, PyTorch
- **Soft Skills:** rigorous analysis, technical communication, independent problem-solving

Backdoor Attack in Large Language Model's Reasoning

Supervisor: Dr. Honghui Xu,

Kennesaw State University, Georgia, USA

July 2026 – Present

- Conducting research to identify the vulnerability of LLM reasoning
- **Technical Skills:** LLM reasoning evaluation (chain-of-thought analysis), backdoor/trigger injection techniques, Python, PyTorch, Hugging Face Transformers
- **Soft Skills:** hypothesis-driven experimentation

WORK EXPERIENCE

Senior Software Engineer

Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

January 2024 – Present

- **Project 1: Cloud Solutions – Medison** Jan 2024 – Feb 2025
 - Contributed to the R&D team of Samsung Medison's Cloud Solutions project
 - Served as a streaming technology specialist to publish the generated stream to the web in real-time
 - Developed an application to remove the dependency on ffmpeg to read and show generated stream, implemented with double buffering using DirectShow and Media Foundation
 - **Technical Skills:** Windows application development, computer networking, real-time transport protocols, C++, C#, Python, JavaScript, Machine Learning, Docker, Cloud solutions using AWS and Azure, FFmpeg, MediaMTX, WebRTC, Libdatachannel, Namedpipe
 - **Soft Skills:** Communication, negotiation, time management, task & responsibility management, teamwork, collaboration, continuous self-improvement
- **Project 2: Cardiovascular Workstation – Medison** Mar 2025 – Mar 2026
 - Implemented the UI of the Server (Windows software for ultrasound machine & website)
 - Engineered backend design

- Architected the database management design
 - Successfully deployed and configured the system in an on-premises environment
 - **Technical Skills:** React, WPF, .NET, networking, IIS Windows Server
 - **Soft Skills:** Teamwork, intern mentoring, leadership, ownership
- **Project 3: Cardiac TEE Rendering & Optimization** April 2026 – Present
- Leading the optimization of UI design for the 5D Viewer
 - Performing key responsibilities for actions on rendered objects
 - **Technical Skills:** C#, C++, HLSL, DirectX
 - **Soft Skills:** Leadership, ownership, initiative, cross-functional collaboration

ACADEMIC PROJECTS

Gardening Tukitaki

University of Dhaka, Dhaka, Bangladesh 2023

- Developed a blog platform for gardeners, agriculturists, farmers and nature admirers for sharing knowledge about gardening, also featuring e-commerce capabilities([Project Code](#))
- Designed and implemented the database architecture, the backend APIs with a user-friendly frontend UI
- **Technical Skills:** Python, Django, HTML, CSS, JavaScript, SQLite3, web-publishing
- **Soft Skills:** Team leading, clean code, code refactoring, planning, logical thinking

Find Your Doctor

University of Dhaka, Dhaka, Bangladesh 2022

- Developed an android app to facilitate users in finding and scheduling appointments with their desired doctors maintaining dual profiles for both medical professionals and patients ([Project Code](#))([Project Documentation](#))
- Contributed to the UX and UI design with backend implementation
- Prepared and analyzed necessary planning and documentation
- **Technical Skills:** Java, Firebase, Android Studio SDK, automated testing
- **Soft Skills:** Functional and non-functional requirement analysis, project planning, team leading, collaboration, smooth presentation, report writing and proper documentation

Book Your Ride

University of Dhaka, Dhaka, Bangladesh 2021

- Engineered an online bus ticket booking system to book bus tickets online and see the schedules of the buses according to destination and date ([Project Code](#))
- Digitalized the system by providing the users the online copies of the tickets
- Built up the back-end architecture in synchronization with the database and frontend UI
- **Technical Skills:** Java, Firebase, Android Studio SDK
- **Soft Skills:** Project planning, team leadership, negotiation

Database Project on Library Management System

University of Dhaka, Dhaka, Bangladesh 2020

- developed a database to manage the library of an institute ([Project Report](#))
- designed the database based on normalization and created views to execute its smooth conduction
- **Technical Skills:** SQL, Oracle Database
- **Soft Skills:** Management system analysis, project planning

ACHIEVEMENTS

- 1st position, Basic AI/ML Exam, Samsung R&D Institute Bangladesh
- Talent Pool Scholarship in HSC, 2019-2023, Dhaka, Bangladesh
- Agriculturist (Krishibid) Institute Scholarship, 2019-2023, Dhaka, Bangladesh
- Perfect Attendance Award, Notre Dame College, 2018, Dhaka, Bangladesh
- World Literature Center (Bishwo Shahitto Kendro) Award, 2017, Dhaka, Bangladesh
- Agriculture (Krishi) Foundation Talent Award, 2016-2017, Dhaka, Bangladesh
- 2nd Position, Notre Dame Annual Science Festival 2014 & 24th GKC, Dhaka, Bangladesh

CO-CURRICULAR ACTIVITIES

Online courses -

- Advanced AI Training | Bangladesh University of Engineering and Technology
- Advanced Python: Python packaging. Pip install your scripts! | Udemy
- Machine Learning & Data Science: The Complete Visual Guide | Udemy
- Executive Briefing: Artificial Intelligence (AI) + ChatGPT | Udemy
- C++ Templates and the STL | LinkedIn Learning
- Microservices: Designing Highly Scalable Systems | Udemy
- How to tell a story with data | Udemy
- Leadership: How to Influence, Inspire and Impact as a Leader | Udemy

EXTRA-CURRICULAR ACTIVITIES

- Volunteer, Code Samurai 2022 by BJIT, University of Dhaka, Dhaka, Bangladesh
- Member, DU YESS, 2019-2023, University of Dhaka, Dhaka, Bangladesh

SKILLS

- **Programming languages:** C, C++, C#, Java, Python, JavaScript
- **Databases:** SQL, SQLite3, Firebase Real-time Database & Cloud Firestore
- **Framework & SDK:** Django, Android SDK, Bootstrap, React
- **Operating System:** Windows, Linux (Debian)
- **Machine Learning Tools:** NumPy, Pandas, scikit-learn
- **Web Languages:** HTML, CSS
- **Other Tools:** Docker, Azure Cloud Computing, WebRTC

REFERENCES

- | | |
|---|---|
| <ul style="list-style-type: none"> • Dr. Upama Kabir
Professor, Department of Computer Science
& Engineering
Dean, Faculty of Engineering & Technology
University of Dhaka
Dhaka, Bangladesh
upama@cse.du.ac.bd | <ul style="list-style-type: none"> • Dr. Mosarrat Jahan
Associate Professor
Department of Computer Science &
Engineering
University of Dhaka
Dhaka, Bangladesh
mosarratjahan@cse.du.ac.bd |
| <ul style="list-style-type: none"> • Dr. Shaily Kabir
Assistant Professor
Computer Science
Faculty of Science
University of Nottingham
Nottingham, United Kingdom
Email: Shaily.Kabir5@nottingham.ac.uk | |